EXPERIMENT 2: LOAD A PRE-TRAINED TRANSFORMER MODEL

Aim

To explore the Hugging Face Transformers library and load a pre-trained Transformer model for sentiment analysis.

Algorithm

1. Import the pipeline function from the Transformers library.
2. Load a pre-trained sentiment analysis model.
3. Provide an input sentence.
4. Pass the sentence to the model.
5. Display the predicted sentiment and confidence score.

Program

```
# Experiment 2: Load a Pre-trained Transformer Model

from transformers import pipeline

# Load the pre-trained sentiment analysis model

classifier = pipeline(

    "sentiment-analysis",

    model="distilbert/distilbert-base-uncased-finetuned-sst-2-english"

)

# Input sentence

text = "Generative AI is very useful for students."

# Perform sentiment analysis

result = classifier(text)
```

```
# Display output

print("Input Text:", text)

print("Predicted Sentiment:", result[0]["label"])

print("Confidence Score:", round(result[0]["score"], 4))
```

Note: The first execution downloads the model, so it may take some time.

Input

Generative AI is very useful for students.

Expected Output

Input Text: Generative AI is very useful for students.

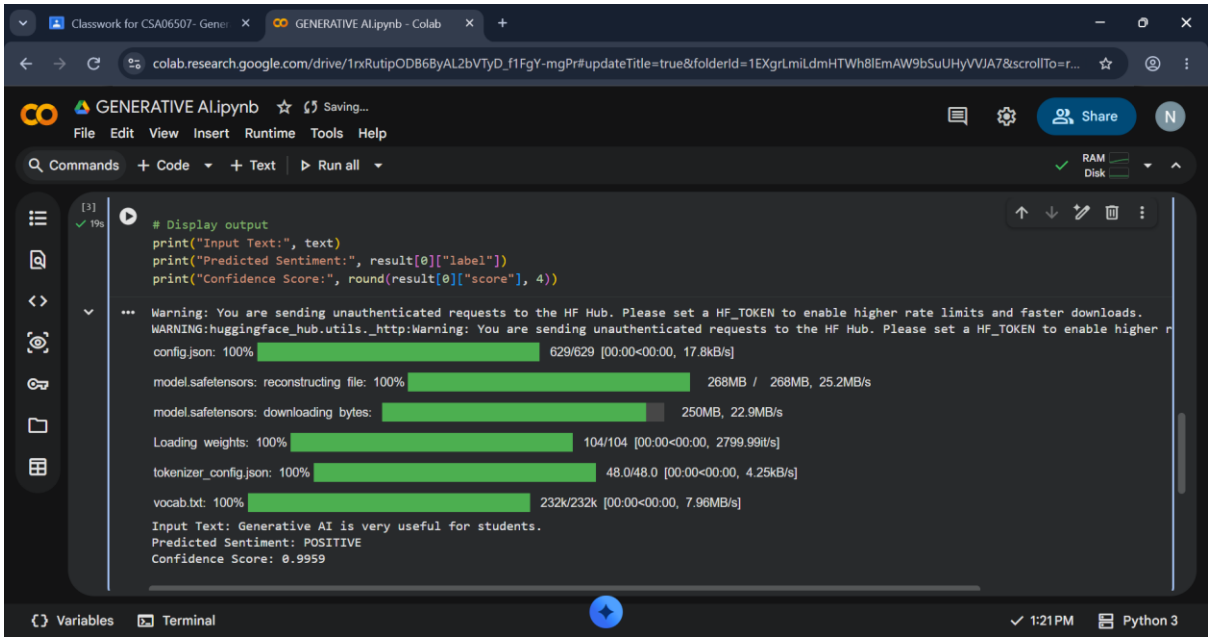
Predicted Sentiment: POSITIVE

Confidence Score: 0.99xx

The confidence score is illustrative and may vary. The model requires an internet connection to download its files the first time.

Explanation

- **Hugging Face:** A platform that provides pre-trained AI models and tools.
- **Transformer:** A neural network architecture used for language understanding and generation.
- **Sentiment analysis:** Identifying whether a text expresses a positive or negative sentiment.
- **Pipeline:** A convenient function that simplifies loading and using pre-trained models.



Result

Thus, a pre-trained Transformer model was loaded using Hugging Face and successfully used to classify the sentiment of the input sentence.